

SARYA, Russian Federation

WMO: 272710

Lat: 58.367N

Lon: 45.533E

Elev: 410

StdP: 14.48

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.5	-19.1	-30.2	1.0	-25.3	-23.6	1.5	-18.9	17.4	19.2	15.6	19.1	2.2	340	0.600

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.4	84.2	68.5	80.6	66.5	77.3	64.7	70.4	80.9	68.5	77.8	66.7	74.7	5.7	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.0	100.8	74.8	65.2	94.7	72.9	63.5	89.0	71.2	34.5	81.2	33.0	77.9	31.6	74.7	78.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.9	13.9	12.2	-32.2	88.3	5.5	3.8	-36.2	91.0	-39.3	93.2	-42.4	95.3	-46.4	98.1		
(5)			-32.4	72.8	5.4	2.6	-36.3	74.7	-39.4	76.2	-42.5	77.7	-46.4	79.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	37.5	10.5	13.2	23.9	39.0	51.0	58.8	63.7	58.7	49.5	38.5	25.6	16.2
	DBStd	21.05	14.92	14.31	9.90	9.50	9.16	7.62	6.66	7.12	7.70	8.34	10.84	14.49
	HDD50	5788	1223	1032	810	349	102	14	1	9	101	366	733	1048
	HDD65	10167	1688	1452	1275	781	441	215	107	221	469	822	1183	1513
	CDD50	1225	0	0	0	18	133	277	425	278	85	9	0	0
	CDD65	130	0	0	0	0	7	28	66	26	3	0	0	0
	CDH74	1527	0	0	0	5	127	342	752	282	19	0	0	0
	CDH80	379	0	0	0	0	23	82	191	81	2	0	0	0
Wind	WSAvg	5.7	5.8	6.1	6.2	6.2	6.2	5.3	4.5	4.8	5.2	6.2	6.1	6.2
Precipitation	PrecAvg	25.4	1.7	1.3	1.3	1.3	2.0	3.4	3.1	2.8	2.1	2.4	2.2	1.9
	PrecMax	34.4	3.3	2.8	3.0	3.5	4.4	6.5	7.4	5.9	4.7	5.2	5.5	3.1
	PrecMin	17.0	0.9	0.5	0.2	0.4	0.7	0.7	0.4	0.9	0.4	0.6	0.2	0.7
	PrecStd	3.7	0.6	0.6	0.7	0.8	0.8	1.5	1.9	1.1	0.9	1.1	1.1	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.9	36.9	50.3	73.0	83.0	87.4	89.4	87.9	76.5	63.7	48.8	41.2
		MCWB	33.8	34.5	40.6	54.6	63.1	69.8	70.1	70.7	62.2	53.5	46.5	40.5
	2%	DB	33.5	34.7	42.7	66.1	77.4	81.8	84.7	80.8	70.7	57.1	43.5	36.8
		MCWB	32.5	33.2	36.2	51.6	60.9	66.6	69.1	67.9	59.7	50.2	42.0	35.7
	5%	DB	32.2	33.2	39.3	60.1	72.5	77.8	81.5	76.2	66.0	53.0	40.3	34.1
		MCWB	31.4	32.1	34.1	48.0	57.5	64.3	67.8	64.8	57.6	49.0	39.0	33.1
	10%	DB	30.0	31.6	36.8	54.7	68.2	74.1	78.2	72.4	62.2	49.7	37.5	32.7
		MCWB	29.1	30.6	33.2	45.4	54.6	62.8	66.2	63.3	55.7	47.1	36.5	31.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.2	35.2	41.2	55.7	66.8	71.7	72.8	72.5	64.3	55.3	46.7	40.8
		MCDB	34.4	36.1	47.6	69.3	79.3	81.7	82.9	84.4	71.4	60.1	48.3	41.3
	2%	WB	32.8	33.5	37.7	52.5	63.0	69.0	70.7	69.3	61.3	51.8	42.5	36.0
		MCDB	33.2	34.4	41.3	63.9	72.7	79.3	80.8	78.7	68.1	55.8	43.1	36.7
	5%	WB	31.4	32.3	35.5	49.5	59.7	66.6	69.2	66.8	58.9	49.5	39.2	33.3
		MCDB	32.1	33.2	38.0	58.5	68.7	74.6	78.8	73.9	64.9	52.2	40.0	33.9
	10%	WB	29.0	30.5	33.7	46.2	56.8	64.3	67.6	64.4	56.3	47.2	36.5	32.1
		MCDB	29.8	31.5	36.1	53.3	65.0	71.5	76.2	70.9	61.3	49.8	37.3	32.7
Mean Daily Temperature Range	5% DB	MDBR	12.4	14.7	17.2	18.5	21.6	20.5	21.4	19.2	16.5	10.7	7.6	9.9
		MCDBR	7.4	8.6	18.2	27.3	28.7	26.8	26.6	25.9	23.0	17.3	8.9	6.5
		MCWBR	7.4	7.9	12.6	14.6	13.6	13.0	12.3	12.8	12.7	11.1	8.3	6.5
	5% WB	MCDBR	7.5	8.1	12.7	25.0	24.2	22.5	23.2	22.2	18.9	14.8	8.3	6.1
		MCWBR	7.6	7.9	9.2	14.3	12.8	12.0	11.6	11.8	10.9	10.2	8.1	6.2
Clear-Sky Solar Irradiance	taub	0.275	0.310	0.345	0.394	0.368	0.366	0.397	0.400	0.357	0.337	0.300	0.263	
	taud	2.057	2.057	2.041	2.129	2.397	2.424	2.350	2.320	2.422	2.353	2.178	2.025	
	Ebn at Noon	167	218	246	255	274	276	263	252	247	214	159	129	
	Edh at Noon	17	28	38	41	34	33	35	34	26	21	15	13	
All-Sky Solar Radiation	RadAvg	131	358	772	1202	1652	1730	1730	1313	767	332	122	73	
	RadStd	22	94	105	141	126	143	163	124	99	56	17	15	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)		+1.01	N/A	N/A	N/A	N/A	N/A	N/A	-373	N/A	N/A

Nomenclature: See separate page